1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.

2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it.

- Here's the overview of main hardware networking devices :

1. Modem

- Connect your home or office network to your Internet Service Provider (ISP) by converting signals between the ISP's network and your local network.
- **Real-Life Example:**
- At home, connecting your house to the internet provided by your ISP.

2. Router

- Directs data between your local devices and the internet, allowing multiple devices to share one internet connection.

- **Real-Life Example:**
- At home, sharing the internet with laptops, phones, tablets, and smart TVs over Wi-Fi.

3. **Switch**

- Connects multiple wired devices within the same local network and sends data only to the intended device.
- **Real-life Example:**
- In an office or school, connecting desktop computer, printers and servers with Ethernet cables.

4. **Access Point (AP)**

- Provides Wi-Fi access to devices by connecting them to a wired network. It extends or creates a wireless network.
- **Real-Life Example:**
- In a school, hotel or large office, providing wireless internet coverage in different rooms or floors.

5. **Ethernet Cable:**

- Physically connects network devices and transfers data through a wired connection.
- **Real-life Example:**
- Connecting a desktop computer to a router for a fast and stable internet connection.

3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.

1. **Ethernet Cable**

- Commonly used to connect computers, routers, switches and other devices in home and office LANs.

- **Link**
  https://commons.wikimedia.org/wiki/File%3AEthernet_cable.JPG?utm_source=chatgpt.com

2. **Coaxial Cable**

- Commonly used for cable TV connections, cable internets and connecting antennas.
- **Link**
  https://commons.wikimedia.org/wiki/File%3ACoax_cable.jpg?utm_source=chatgpt.com

3. **Fiber Optic Cable**

- Commonly used for high-speed internet backbones, data centers, and long-distance communication networks.
- **Link**
  https://en.wikipedia.org/wiki/Fiber-optic_cable?utm_source=chatgpt.com

4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.

Feature	**Router**	**Switch**
Main use	Connects different networks (e.g. home network to the internet)	Connect devices within the network(e.g. computers, servers, printers)
Typical location	At the gateway of a network, often between a local network and the ISP.	Inside a LAN such as an office or home network.
Number of networks can connect	Can connect multiple separate networks.	Typically connects multiple devices on a single network.
Internet sharing	Can share an Internet connection among multiple devices.	Usually does not provide Internet routing by itself.